MODELLING CLOUDS IN PLANETARY ATMOSPHERE

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DEDICATION

To the Almighty, my father and most importantly to ***life*** *for teaching me how to be a human ...*

DECLARATION

I hereby declare that I am the sole author of this thesis in partial fulfillment of the requirements for a postgraduate degree from National Institute of Science Education and Research (NISER). I authorize NISER to lend this thesis to other institutions or individuals for the purpose of scholarly research.

Signature of the Student
Date: 19th May 2026

The thesis work reported in the thesis entitled **Modelling Clouds in Planetary Atmosphere** was carried out under my supervision, in the school of *Physical Sciences* at NISER, Bhubaneswar, India.

Signature of Thesis Supervisor

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School : Earth and Planetary Sciences

Signature of the thesis supervisor
School: Physical Sciences
Date: 19th May 2026

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ABSTRACT

The clouds in Jupiter's atmosphere are predominantly composed of ammonia (NH_3), as the local temperature–pressure conditions favor its condensation. In this work, I implement both analytical and numerical cloud models to calculate the vertical profiles of condensate abundance, particle size distribution, and number density for cloud species in planetary atmospheres. The methodology closely follows the framework of Ackerman & Marley (2001) [1], using a sedimentation efficiency parameter (f_{sed}) to describe the balance between upward turbulent mixing and downward gravitational settling. For Jovian conditions, I compute ammonia cloud profiles for different values of f_{sed} and compare the resulting condensate structures with established cloud model **Virga**, thereby examining how cloud particle properties and vertical distributions vary across atmospheric pressure levels.

Building on this framework, I extend the model to hot exoplanet atmospheres, with particular focus on WASP-17b, where high atmospheric temperatures allow the condensation of refractory species such as silicate clouds. Motivated by recent observational studies suggesting the presence of SiO_2 cloud opacity in the atmosphere of WASP-17b, I implement SiO_2 condensation curves and compute cloud profiles using both constant-K_{zz} and pressure-dependent eddy diffusion approaches. This time I have also employed the *initial value problem* solver method **RK23**. This study provides a unified framework for modeling both volatile clouds in Solar System giant planets and refractory clouds in highly irradiated exoplanets, and serves as a foundation for future cloud formation simulations within our group's adaptive planetary atmosphere model, **SANSAR**.

Contents

List of Figures

Chapter 1

Introduction

1.1 Motivation

Clouds are among the most significant sources of uncertainty in the characterization of planetary atmospheres. Their vertical structure, particle size distribution, and condensate abundance directly shape the observed spectra of both Solar System planets and exoplanets, making accurate cloud modeling an essential component of any self-consistent atmospheric retrieval or forward modeling framework. While mature cloud modeling tools such as Virga exist, they are designed as standalone packages with fixed internal assumptions and interfaces. Integrating such tools into a coupled atmospheric model — where the cloud module must exchange temperature, pressure, chemical abundance, and radiative flux information dynamically with other physical components — is non-trivial and often impractical. Our group is developing SANSAR, an adaptive planetary atmosphere model that requires a fully native, modular cloud formation component whose assumptions, microphysical parameters, and condensation chemistry can be modified and extended as the model evolves. The present work is therefore motivated by the need to build this cloud module from first principles, implementing the Ackerman & Marley (2001) framework in a form that is transparent, controllable, and directly integrable into SANSAR. By constructing the model from scratch — including the vertical transport equation, particle fall-speed calculations across drag regimes, and the condensate selection framework — we retain complete control over the physical assumptions at every stage. Virga serves not as a replacement but as a validation benchmark, allowing us to assess the correctness of our implementation before coupling it to the broader atmospheric model. A second motivation is generality across atmospheric regimes. Clouds in Jupiter's atmosphere are composed of volatile condensates such as ammonia forming at relatively cool temperatures, whereas highly irradiated exoplanets like WASP-17b host refractory silicate clouds forming at temperatures exceeding 1000 K. Building a single unified framework capable of handling both regimes — with the same transport equation, particle microphysics, and condensate stability analysis, differing only in input chemistry and thermodynamic data — provides a physically consistent foundation for future atmospheric studies across the full diversity of planetary environments our group plans to model with SANSAR.

1.2 Background

Clouds play a fundamental role in regulating the radiative, chemical, and dynamical structure of planetary atmospheres. By absorbing and scattering radiation, they shape both reflected and emitted spectra, thereby influencing a planet's albedo, thermal balance, and observational appearance. In both Solar System planets and exoplanets, the presence of clouds introduces significant uncertainties in atmospheric characterization, since the vertical location, thickness, and particle properties of cloud layers strongly affect the interpretation of observed spectra. Understanding cloud formation and vertical cloud structure is therefore essential for connecting atmospheric models with spectroscopic observations.

In Jupiter, the uppermost major cloud deck is formed by ammonia (NH_3), which condenses when the local temperature–pressure profile intersects the ammonia saturation curve, typically at pressures of order 0.1–0.7 bar depending on the assumed atmospheric profile as currently we are benchmarking our model with Virga. The resulting cloud structure is determined by the competition between condensation, turbulent mixing, and particle sedimentation. A widely used one-dimensional description of this balance is provided by the cloud model of Ackerman & Marley (2001) [1], which introduces the sedimentation efficiency parameter f_{sed} (also written as f_{rain} in earlier formulations). This parameter measures the relative importance of downward particle settling compared to upward turbulent transport, and allows the model to represent a range of cloud morphologies—from vertically extended cloud decks with small particles to compact clouds with larger, rapidly settling condensates.

In this work, I implement the Ackerman & Marley (2001) [1] framework using both analytical and numerical approaches to compute vertical profiles of condensate abundance, effective particle size, and particle number density. As a baseline validation case, I first model ammonia cloud formation in Jupiter's atmosphere for multiple values of f_{sed}, and compare the resulting cloud structures with established eddysed-style and Virga-based implementations. This enables a systematic study of how sedimentation efficiency controls cloud thickness, cloud-base location, and particle growth under Jovian conditions.

Building on this Solar System benchmark, I then extend the same framework to the highly irradiated exoplanet WASP-17b, where the atmospheric temperatures are sufficiently high for refractory condensates such as silicates to form. The silicate clouds form at the pressure levels of 10^{-5} bar and 10^{-6} bar [7]. Motivated by recent observational evidence suggesting the presence of SiO_2 cloud opacity in the atmosphere of WASP-17b, I implement SiO_2 condensation curves and compute cloud profiles using constant-K_{zz}. By comparing the resulting condensate mass mixing ratio, effective radius, and number density profiles with outputs from the Virga cloud model, I assess the consistency of the numerical implementation and explore the dependence of cloud structure on sedimentation efficiency, metallicity, and atmospheric thermal profile. This work therefore establishes a unified cloud-modeling framework applicable to both volatile clouds in Solar System giant planets and refractory clouds in hot exoplanets, and provides a foundation for future cloud microphysics studies within our group's adaptive planetary atmosphere model, **SANSAR**.

Chapter 2

Theory and Formulations

2.1 The fundamentals of cloud formation

In meteorology, clouds are suspensions of liquid droplets, ice particles, condensate aerosols, or other particles suspended in the atmosphere of a planetary body or similar space. Water or various other chemicals may be comprising as crystals and droplets, e.g.- ammonia clouds on Jupiter and water vapor clouds on Earth.

In planetary atmospheres—especially giant planets such as Jupiter and hot Jupiter planets like WASP -17b ;clouds exert a dominant control on radiative transfer, atmospheric circulation, thermal structure, and observable spectra. Their formation and evolution arise from a balance between thermodynamics, micro-physical growth processes, and turbulent transport.

Cloud microphysics deals with the small-scale physical processes that govern the formation, growth, and movement of cloud particles. These include condensation, nucleation, evaporation, sedimentation, and turbulent mixing. Although these processes occur on microscopic scales, their cumulative effects determine large-scale cloud structure and optical properties.

2.1.1 Cloud Formation

Air Parcel and Turbulence

In any planetary atmosphere, molecular mixing of gases and substances takes place only up to a few centimeters in the atmosphere. To talk about mixing of substances across the vertical coordinate, in macroscale, the concept of air parcels is important.

Each air parcel is assumed to be :

i) thermodynamically insulated from its environment, and its temperature changes adiabatically as it rises or sinks and always remains at the same pressure level as the environmental air in the surroundings, which is assumed to be in hydrostatic equilibrium.

ii) moving slowly enough that the kinetic energy of the air parcel is at a negligible scale of the total energy.

[4] When this warm air parcel laden with the gaseous form of elements and chemical species rises, it expands and cools. If the chemical species cools to a temperature where it can condense, a condensation cloud will form. The moment the temperature of the air parcel is equal to the **dew point temperature** (where relative humidity is 100 %) of the species, the cloud base is formed. Now the parcel of air will continue rising up to high altitude levels until the surrounding atmospheric air becomes cooler than the air parcel, and that is where the cloud formation stops. This vertical length till which the cloud is formed is called the height of the cloud. A species can only start to condense when the partial pressure of the species vapor exceeds the partial pressure of the gaseous species when air is fully saturated. For any gaseous species that might condensate, [4]

$$P_g(T) = X_g P(T) \geq P_{Vg}(T)$$

where P(T) is the total gas pressure,P_g is the gas partial pressure, X_g is the gas mixing ratio and P_{Vg} is the saturation vapor pressure.

The cloud base is at the intersection of the planet's temperature-pressure profile with the saturation vapor pressure curve. This curve is the equilibrium curve between the gas and liquid or solid state of the species.

The saturation vapor pressure as a unction of temperature is known as the Clausius Clapeyron Curve - [4]

$$\frac{dP_V}{dT} = \frac{L}{T(V_2 - V_1)}$$

Where V = 1/ ρ and is known as the specific volume and 1,2 subscript refers to the transition of state of species from liquid or solid to gaseous phase. If we consider the ideal gas then the specific volume of solid or liquid is small compared to the vapor phase, i.e $(V_2 - V_1) \sim V_1 = R_v T/P$ where R_v is the specific gas constant for the vapor in the planet. Then the above equation becomes :

$$\frac{dP_v}{dT} = \frac{LP_v}{R_v T^2}$$

here L is latent heat of vaporisation. As the air parcel rises it expands and cools adiabatically until $P_v = P_{sat}$. Beyond this point he air becomes supersaturated, and condensation begins. In many cloud models (here, [1] the assumption is made that supersaturation is negligible due to abundance of condensation nuclei. Thus,

$$q_s = \frac{R_d}{R_v}\frac{P_{sat}}{P - P_{sat}}$$

where P is the total pressure and R_d is mass of dry air in the planet. Now, for clouds to form in a large scale height and be broad, only molecular diffusion is not enough. We need the intervention of turbulence. The large scale mixing is mostly facilitated by the set up of convective cycles or convective current in which the hot air parcel rises passing the dew point temperature and beginning the formation of clouds the hot air parcel rises and cools down and while the heavier, colder condensed species is coming down it is again getting re-evaporated due to the surrounding atmospheric temperature being warmer than it and thus a cloud is sustained. To sustain a cloud, molecular mixing, turbulence mixing and convection current must always be an active process . Only when the condensed species is way cooler than the surrounding atmospheric layer do we see precipitation in form of rain, snow or virga. Turbulence continuously mixes vapor and condensate vertically. Vertical mixing is often parameterized as an eddy diffusion process:

$$F_{mix} = -K_{zz} \frac{dq_t}{dz}$$

here, K_{zz} is the 1-D eddy diffusion coefficient along the vertical coordinate

q_t is the total mass mixing ratio (vapor (q_v) + condensate (q_c)). This term behaves familiar to **Fick's first law of diffusion** transporting vapor upward from deep levels and lifting condensate upward against gravity.

In the convective regions, K_{zz} is derived from the mixing length theory. **A**

Sedimentation (Rainout)

Sedimentation, or rainout, is one of the central processes controlling the vertical distribution of condensates in planetary atmospheres. After condensation occurs, cloud particles experience gravitational settling and drift toward deeper atmospheric layers. This downward flux of condensate competes with upward turbulent transport, and the balance between the two determines whether a cloud remains vertically extended or becomes confined to a narrow layer near its cloud base.

In the Ackerman & Marley (2001) [1] model, this competition is captured by the sedimentation efficiency parameter f_{sed}, defined as the ratio of the mass-weighted mean settling velocity of particles to the characteristic upward mixing velocity. Thus, f_{sed} provides a simple but physically meaningful way to describe cloud morphology. Lower values of f_{sed} correspond to weaker rainout and more vertically extended clouds with smaller particles, whereas higher values of f_{sed} imply stronger rainout, larger particles, and thinner cloud decks.

The settling velocity of a particle depends on its radius, the local atmospheric density and viscosity, and the drag regime. In the low-Reynolds-number limit, the terminal velocity is described by the Stokes expression

$$v_{\mathrm{fall}} = \frac{2}{9}\,\frac{\beta(\rho_p - \rho_a)\,g\,r^2}{\eta},$$

where η is the dynamic viscosity of the atmosphere, g is the gravitational acceleration, ρ_p is the condensate density, ρ_a is the atmospheric density, r is the particle radius, and β is the Cunningham slip correction factor. For larger particles, where the Reynolds number increases and the assumptions of Stokes flow break down, the terminal velocity must instead be computed using more general drag prescriptions or empirical drag laws. Since the fall velocity increases with particle size, sedimentation naturally promotes the preferential removal of larger particles from the upper atmosphere. A detailed derivation of the fall velocity and drag-regime transitions is presented in Section **B**.

2.1.2 The main Equation

The cloud microphysics framework developed by [1] provides a one-dimensional steady-state description of cloud formation in planetary atmospheres, where the upward transport of condensable material by turbulent mixing is balanced by the downward sedimentation of cloud particles under gravity. Despite its relative simplicity, the model successfully captures the dominant physical processes controlling the vertical structure of condensate clouds and has become one of the foundational approaches used in both brown dwarf and exoplanet atmosphere studies.

In this framework, the total condensable material consists of both vapor and condensate components, represented through the total mixing ratio,

$$q_t = q_v + q_c,$$

where:

- q_v is the vapor mixing ratio,
- q_c is the condensate mixing ratio.

The steady-state transport equation balancing turbulent diffusion and sedimentation is written as:

$$-K_{zz}\frac{\partial q_t}{\partial z} - f_{\text{rain}}\, w_*\, q_c = 0 \tag{2.1}$$

where:

- K_{zz} is the vertical eddy diffusion coefficient describing turbulent mixing,
- $\dfrac{\partial q_t}{\partial z}$ is the vertical gradient of the total condensable mixing ratio,
- f_{rain} (or equivalently f_{sed}) is the sedimentation efficiency parameter,
- w_* is the convective velocity scale,
- q_c is the condensate mass mixing ratio.

Physically, the first term in Eq. (2.1) represents upward transport due to atmospheric turbulence and convection, while the second term represents the downward settling of condensate particles caused by gravity. The balance between these two competing processes determines the vertical extent, particle size distribution, and optical thickness of the cloud layer.

The sedimentation parameter f_{rain} is particularly important because it controls how efficiently condensate particles fall relative to the strength of atmospheric mixing. It is defined approximately as the ratio between the characteristic particle settling velocity and the convective velocity scale:

$$f_{\text{rain}} \sim \frac{v_{\text{fall}}}{w_*} \tag{2.2}$$

where:

- v_{fall} is the terminal fall velocity of cloud particles,
- w_* characterizes the upward convective transport.

Larger values of f_{rain} correspond to more efficient sedimentation, producing larger particles and vertically compact cloud decks, whereas smaller values of f_{rain} lead to smaller particles and more vertically extended clouds.

The convective velocity scale is related to the eddy diffusion coefficient through the atmospheric mixing length L:

$$w_* = \frac{K_{zz}}{L} \tag{2.3}$$

where L is typically taken to be comparable to the atmospheric scale height.

Cloud formation begins at the atmospheric level where the vapor mixing ratio becomes equal to the saturation vapor mixing ratio:

$$q_v = q_s. \tag{2.4}$$

This level defines the cloud base. Above this point, the atmosphere becomes supersaturated and excess vapor condenses into cloud particles.

The modern open-source cloud model Virga [5] adopts the same underlying physical philosophy as the Ackerman–Marley framework, but reformulates the transport equation as a continuous ordinary differential equation (ODE) suitable for numerical integration. In Virga, the cloud transport equation is solved as an initial value problem:

$$\frac{dq_t}{dz} = \frac{f_{\rm sed}}{L} \max\left(0, q_t - q_{vs}\right) \tag{2.5}$$

where:

- q_{vs} is the saturation vapor mixing ratio,
- L is the mixing length scale,
- the $\max(0, q_t - q_{vs})$ term ensures that condensation only occurs when the atmosphere becomes supersaturated.

Unlike the original layer-by-layer implementation of Ackerman & Marley (2001), the Virga formulation treats the cloud profile as a continuous differential problem and solves it numerically using modern ODE integration techniques. This allows smoother vertical cloud profiles, improved numerical stability, and easier extension to a wide variety of condensate species and exoplanetary atmospheres.

In this work, we reconstruct the essential physical framework of the Ackerman–Marley model in Python and compare our results directly with the Virga cloud model. The comparison allows us to test the numerical stability, cloud-base location, condensate distribution, particle sizes, and number density profiles obtained from our implementation against a more modern and extensively developed cloud microphysics framework.

For detailed analytical derivations of the governing equations and particle-size relations, refer to Appendix **B**.

2.2 Method

Approach to the Modeling

We are using python interface to model this problem. We first created the pressure temperature layers by making a grid and named it function layer. The function eddysed in the code loops through the gas condensate ammonia (we will options to add more gases to the analysis later) while calling the function of layer. Then it calls the function $calc_{qc}$ to each of the layers of (P,T)of the data and calculates some basic quantities like mean free path, number density , atmospheric viscosity which are required for the calculations of the droplet size radius of the particles of condensate species. In the function $calc_qc$ we start by computing the saturation vapor pressure P_{sat} at a given layer i that has its own corresponding pressure and temperature $(T_i and P_i)$. It is computed through the formula given in **A**. Then it is used to calculate the saturation vapor mass mixing ratio through the formula

$$q_s = \frac{f_s P_{sat}}{R_c T_i \rho_a}$$

where f_s =S+1 is the super saturation which is assumed to be 1 in this model (i.e., S is user defined as 0 it corresponds to potential supersaturation prior to condensation) and R_c is the universal gas constant.q_v (vapor mass mixing ratio)is then compared to q_t in the immediately preceding layer q_{below}. If $q_{below} > q_v$ then we proceed to solve 2.5 numerically through the methods. However to analytically solve the equation 2.5 is the form of :Refer to B

$$q_{t,i} = q_{c,i} + (q_{below} - q_{v,i}) exp(-f_{sed} * \Delta z/L) \tag{2.6}$$

Here, L is the mixing length whose calculation is described in the **A**. This method is the numerical method and here q_v is assumed to be constant throughout the atmospheric layers. In reality it is not the case and we will be taking q_v from equilibrium chemistry in the later stages of the development of the model. We then proceed to calculate the particle fall radius v_{fall} from $calc_{qc}$ and we tried to use the **root finding algorithm brentq** Method (see **A** for definitions) to find the values for the fall radius (r_w) of the particle. f_{rain} is a user defined parameter and we have used it to benchmark our model the figure 1. of [1]). We stress on the fact that $w_* = K_{zz}/L$ and the method we used to calculated K_{zz} is detailed in the paper of [6]. More details in the (**A**).

From the particle fall radius r_w we want to obtain the geometric mean radius r_g via the following equation :

$$r_{g,i} = r_{w,i} f_{sed}^{1/\alpha_1} exp(-\frac{\alpha_i + 6}{2} ln^2 \sigma_g) \tag{2.7}$$

with the associated column droplet number concentration (cm^{-2}) :

$$N_{dz_i} = \frac{3\rho_{a,i} q_{c,i}}{4\pi \rho_p r_{g,i}^3} exp(-\frac{9}{2} ln^2 \sigma_g) dz_i \tag{2.8}$$

where α is the power-law dependence for the particle fall-speed about the solution to the following equation :

$$v_f(r_w) - w_* = 0 \tag{2.9}$$

The fall particle radius is that radius which is computed by finding the radius which balances of the fall speed of the particle (assumed to be spherical here) and the convective velocity scale.

Calculation of the mass mixing ratios

Here q_c , the condensate mixing ratio is calculated using the formula $q_c = q_t - q_s$ and not $q_c = q_t - q_v$ because here, when the cloud starts to form the value of the q_v is capped to q_s as q_s is the maximum vapor mixing ratio that can be held in equilibrium with the condensate at local T,P. To make it more simple , here is how q_t, q_v, q_s, q_c play their roles throughout the atmosphere

1. **Below base of the cloud** $q_v = q_t, q_c = 0, q_v < q_s$

2. **At cloud base** (condensation starts) $q_v = q_s, q_t = q_s, q_c = 0$

3. **Above cloud base** $q_v \to q_s, q_c = q_t - q_s$

Here the *Lewis cloud model (1969)* [2] assumption is made and we are starting below the cloud base where $q_c = 0$ and $q_v = q_{below}$ and condensation of all vapor in excess of saturation is happening at each consecutive layer upward. [1] Thus,

$$q_c(z) = max[0, q_v(z - \Delta z) - q_s(z)] \tag{2.10}$$

$$q_v(z) = min[q_v(z - \Delta z), q_s(z)] \tag{2.11}$$

2.3 Physical Intuition behind the equations

This one- dimensional Eulerian framework of [1] talks about the balance of the two opposite forces - the convective turbulent mixing force acting upwards while maintaining a constant vapor mixing ratio below the cloud and the sedimentation force bringing the condensate particle downward due to action of gravity. Below the cloud base air is too warm and thus saturation vapor pressure is large and so vapor cannot reach saturation hence we have no condensation. At the cloud base vapor mixing ratio equals saturation and condensation begins. Above the cloud base The air becomes supersaturated unless condensation removes vapor and because condensation happens fast relative to atmospheric mixing thus $q_v \to q_s$ (quasi saturated) and excess vapor condenses, q_c increases with increase in altitude. It justifies one of the assumptions that supersaturation is 0 in this model and hence $q_v = q_s$ as the atmosphere is assumed to be nearly saturated . Allowing the supersaturation that persists even after condensation is accounted for represents conditions in which condensation is too slow or there are significant barriers to droplet formation.

Effect of supersaturation :

1. Below cloud base : Pressure and Temperature is high and thus saturation vapor mixing ratio is large if the supersaturation is increased then the threshold rises and the atmosphere can remain unsaturated for longer period of time, thereby pushing the cloud base upward.

2. At cloud base : when the total mixing ratio is equal to the saturation mixing ratio so when the supersaturation is increased then cloud starts forming at higher altitudes and cooler temperatures.

3. Above cloud base : the total mass mixing ratio exceeds the saturation mixing ratio due to condensation and if supersaturation is increased then q_c becomes smaller, clouds become less denser and droplet sizes become smaller and the sedimentation slows down due smaller droplet size, thereby giving a more extended yet less opaque cloud.

Effect of w_* :

1. Larger value of w_* leads to faster mixing, leading turbulent mixing to dominate

2. Smaller value of w_* leads to slower mixing, leading sedimentation to dominate

Effect of f_{rain} :

1. smaller value of f_{rain} = vertically thick clouds, extending through atmosphere leading upward transport dominate

2. Larger value of f_{rain} = thinner depleting clouds leading sedimentation dominate

Effect of metallicity :

1. Smaller value of mh : The gas condenses at a higher pressure , condensation happens at lower temperature as saturation happens with some difficulty.

2. Larger value of mh : The gas condenses at a much lower pressure, saturation point is reached very easily and thus condensation happens at a much higher temperature. Thereby aiding to the formation of the cloud. Cloud formed is also of more quantity and more the metallicity, more is the mass of the species available for condensation.

Effect of K_{zz} :

1. Smaller value of $K_z z$: It makes clouds closer to the condensation level and makes it appear more collected and compact.

2. Larger value of K_{zz} : It makes the cloud more extended above the cloud base and makes the cloud appear more broad and spread out.

3. Higher the value of eddy diffusion parameter, higher is the droplet radius.

4. Higher the value of eddy diffusion parameter, lower is the column density.

Chapter 3

Results and Discussion

3.1 Gas Suggestion Model

In this section of the model, I plotted the condensation curves of all the species against the given PT profile of the planet. The PT curve gets intersected by the species indicating that the selected species which intersect the PT Curve can be found on the planet's atmosphere. The intersection between the atmospheric pressure–temperature (P–T) profile and a condensate's saturation vapor pressure curve marks the level where the vapor reaches saturation, i.e.

$$q_v = q_s,$$

and condensation becomes thermodynamically favorable. This point defines the **cloud base** in a one-dimensional equilibrium cloud model. Below this level, the species remains predominantly in vapor form, whereas above it the excess vapor condenses into cloud particles. The resulting vertical distribution of condensate is then governed by the competing effects of turbulent mixing, sedimentation, and microphysical particle growth.

For the two cases considered here, the interpretation is straightforward. In the case of NH_3, the atmospheric P–T profile intersects the ammonia saturation vapor pressure curve in the cool outer atmosphere of Jupiter. This intersection therefore identifies the **ammonia cloud base**, consistent with the observed upper ammonia cloud deck in Jupiter's atmosphere. In contrast, for SiO_2, the atmospheric P–T profile intersects the silicate saturation curve only at much higher temperatures and deeper pressure levels. This indicates that silicate condensation occurs only in hot, deep atmospheric layers, making such clouds relevant for hot exoplanets such as **WASP-17b**, rather than for Jupiter.

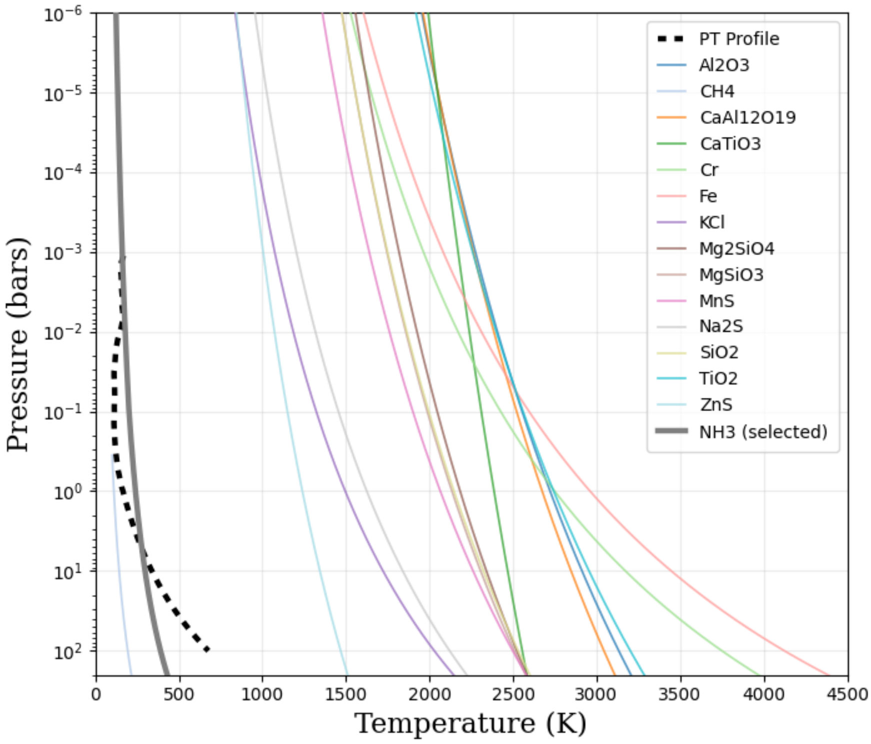

Figure 3.1: Pressure vs Temperature for Voyager profile plotted along with condensation curves of all species with the bold line being that of Ammonia ; indicating that it was selected for the run

A useful physical interpretation is to think of the saturation vapor pressure curve as representing the *maximum amount of vapor the atmosphere can support at a given temperature*, while the atmospheric P–T profile describes the *actual thermodynamic state of the atmosphere*. When the atmospheric profile crosses the saturation curve, the gas can no longer remain entirely in the vapor phase. The excess vapor must then condense, marking the onset of cloud formation.

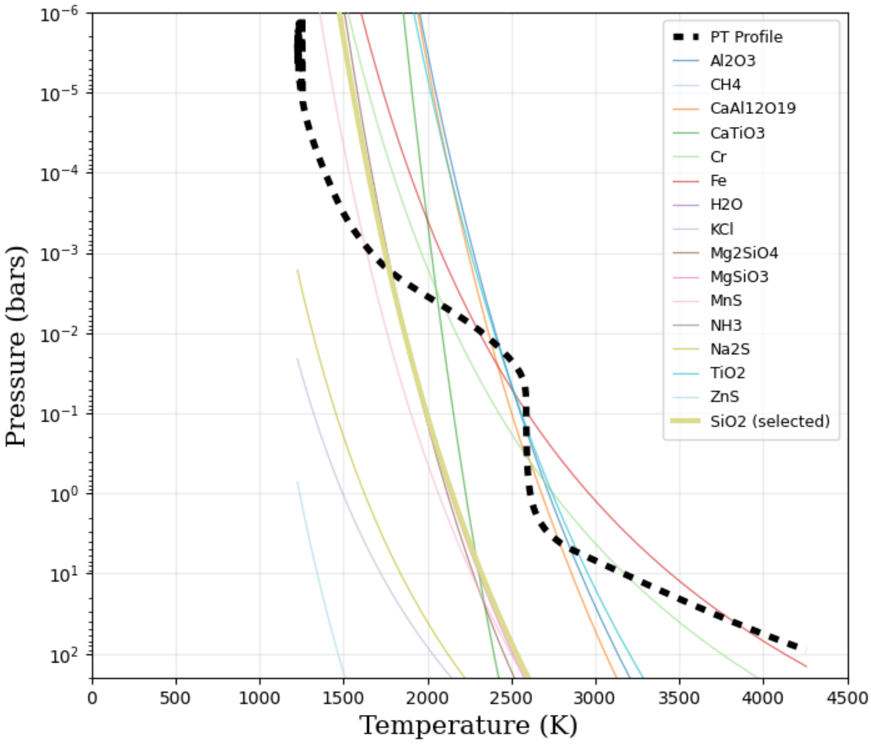

Figure 3.2: Pressure vs Temperature for JWST MIRI profile plotted along with condensation curves of all species with the bold line being that of Silicon Dioxide ; indicating that it was selected for the run

3.2 Effect of K_{zz}

Testing with Jupiter's and WASP -17b PT Profiles.

I have benchmarked my updated model with Virga on two conditions :

1. Constant $K_{zz} = 10^9$

2. Variable k_{ZZ} calculated through the help of convective heat flux and latent heat. Formula given in A [6]

3.2.1 Constant K_{zz}

The value of K_{zz} chosen for this plot was 10^9. The cloud base from Virga simulation came at 0.59 bar and from my model came at 0.59 bar so there is an exact overlap with the VIRGA Results. The graphs for number density and effective droplet radius is also given.

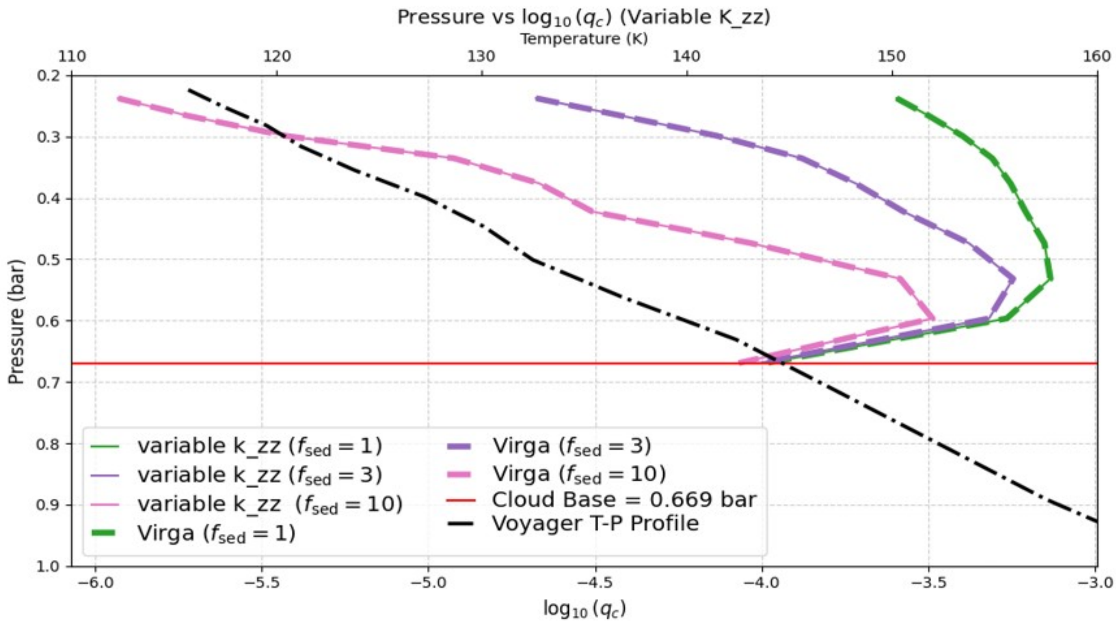

Figure 3.3: Pressure vs $log_{10}q_c$ (constant K_{zz}) for various f_{rain} values and compared with output of Virga for f_{sed} = 1,3,10

Graph of Pressure vs Effective Droplet Radius The graph shows that for f_{sed} 1 and 3 the value of r_{eff} is very close to what the Virga simulations gave while for the value of $f_{sed} = 10$ we see a difference of approximately 20 microns from Virga, indicating that the effective radius of particles as simulated from Virga are bigger than what is simulated from my model. Indicating that the particle size and volume simulated from Virga are bigger than what my code simulation has ran for. The graph shows that particles with higher f_{sed} values have higher effective droplet radius as compared to the lower ones.

Graph of Pressure vs Column density The graph shows a very good agreement of Column density of the particles between my simulation and Virga for the f_{sed} value of 10 with closely following argument from $f_{sed} = 3\&10$. The graph shows

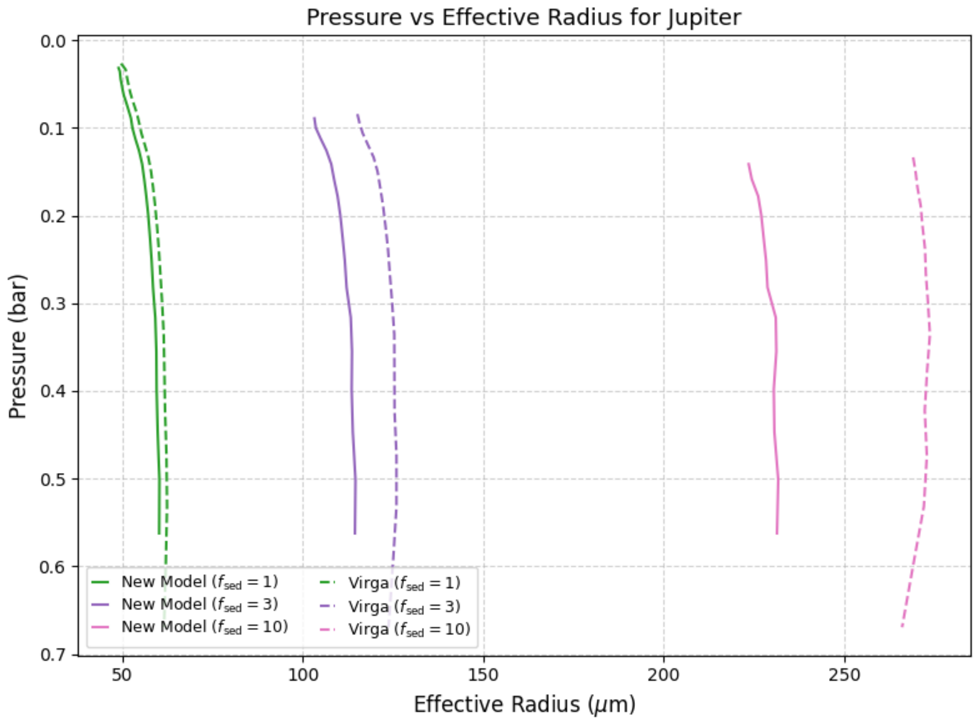

Figure 3.4: Pressure vs effective droplet radius in microns for various f_{rain} values and compared with output of Virga for f_{sed} = 1,3,10

that with higher f_{sed} values the column density of the particles as in the number of particles per cm^2 is far less as compared to one with the lower f_{sed} value. Indicating that particles with smaller droplet radius are more available in the atmosphere than that of the larger droplet radius. The number of particles per unit area increases as we come down towards the surface fo the low f_{sed} value particles and in case of higher f_{sed} value particles the number of particles per unit area remains mostly constant. Indicating a uniform distribution across the pressure layers of the atmosphere.

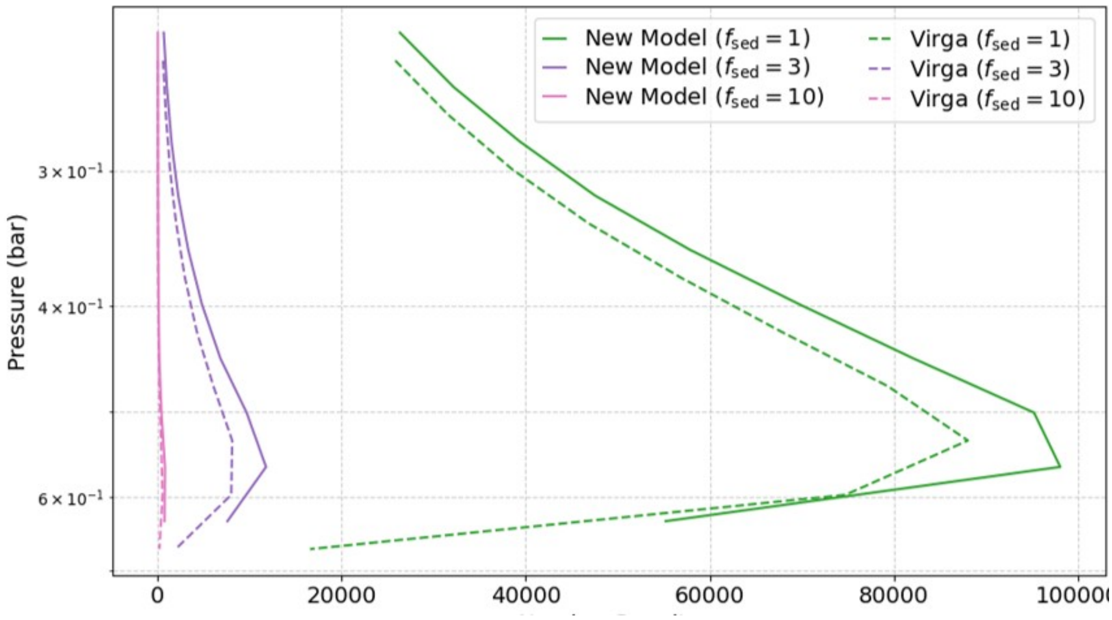

Figure 3.5: Pressure vs column density for various f_{rain} values and compared with output of Virga for f_{sed} = 1,3,10

Reason for the Anomaly

Currently we are using the power law approximation method as [1]:

$$v_f(r) = w_* \left(\frac{r}{r_w}\right)^\alpha$$

$$\alpha = \begin{cases} \dfrac{\log(w_*/v_f)}{\log \sigma_g}, & f_{\text{sed}} > 1, \\ \dfrac{\log(v_f/w_*)}{\log \sigma_g}, & f_{\text{sed}} \leq 1. \end{cases}$$

However VIRGA obtains a better fit for calculation α using *SciPy.optimize* function called *curve-fit* This is causing a different $v_f all$ calculation which is resulting in a **difference in droplet radius and column density**

How K_{zz} changes clouds

Here I used the PT profile of WASP-17b to test for Variable K_{zz} and found that the Variable K_{zz} does play a significant role in the cloud formation. I have varied K_{zz} from 10^8 to 10^{12}. It does not change the quantitative structures of the clouds.

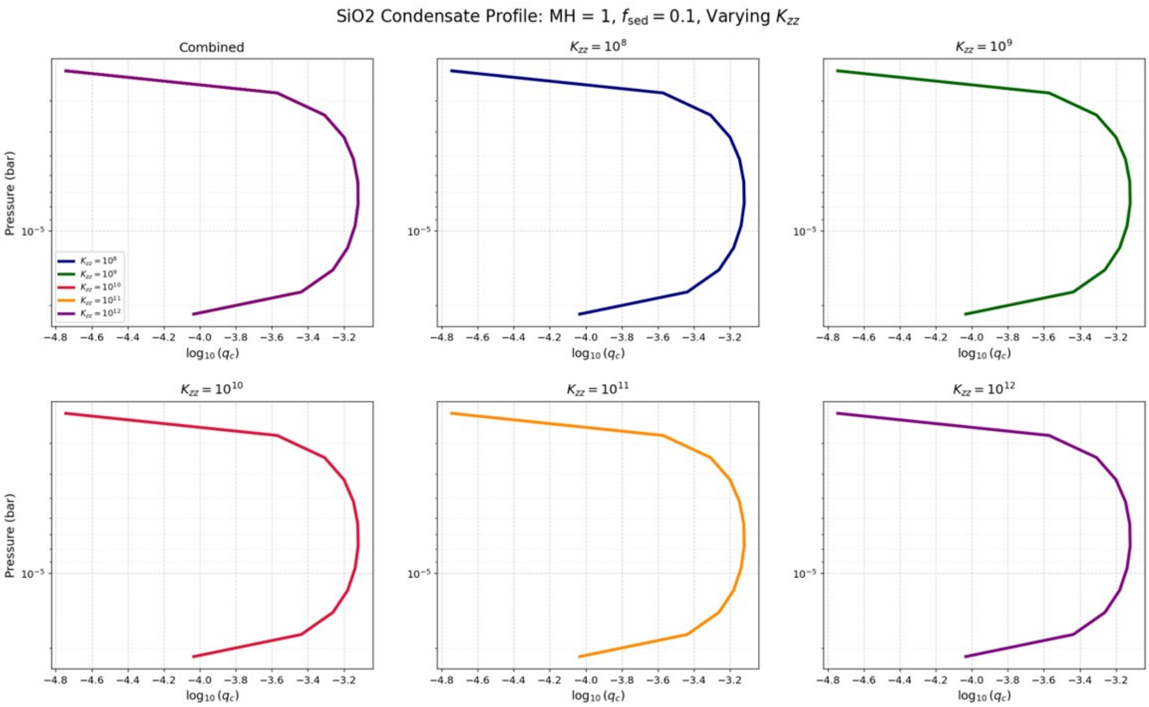

Figure 3.6: Changing the values of K_{zz} to show that it does not change q_c for different values, thereby keeping the cloud location and structure same

The K_{zz} changes the column density and effective radius of cloud particles as seen from the graphs. Thereby it changes the qualitative aspects of the cloud.

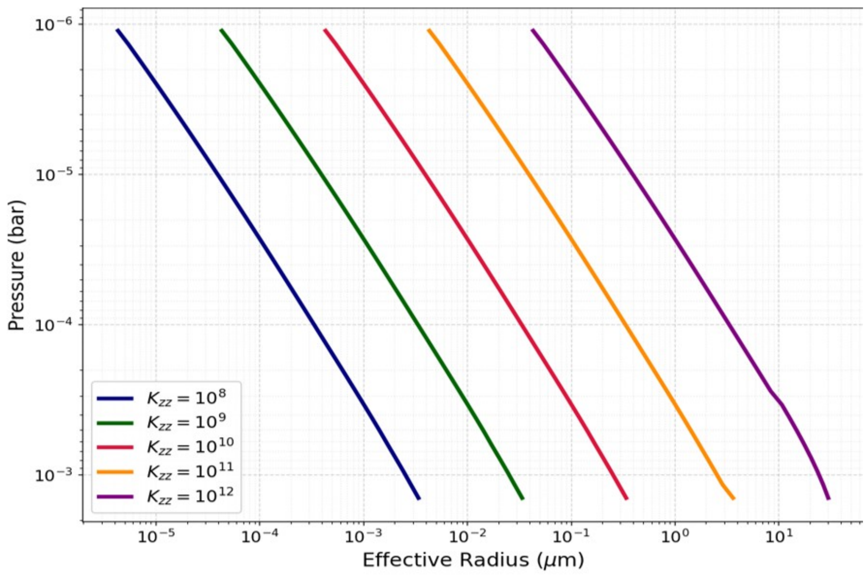

Figure 3.7: For varying K_{zz} we plot Pressure vs droplet radius and seeing the relation.

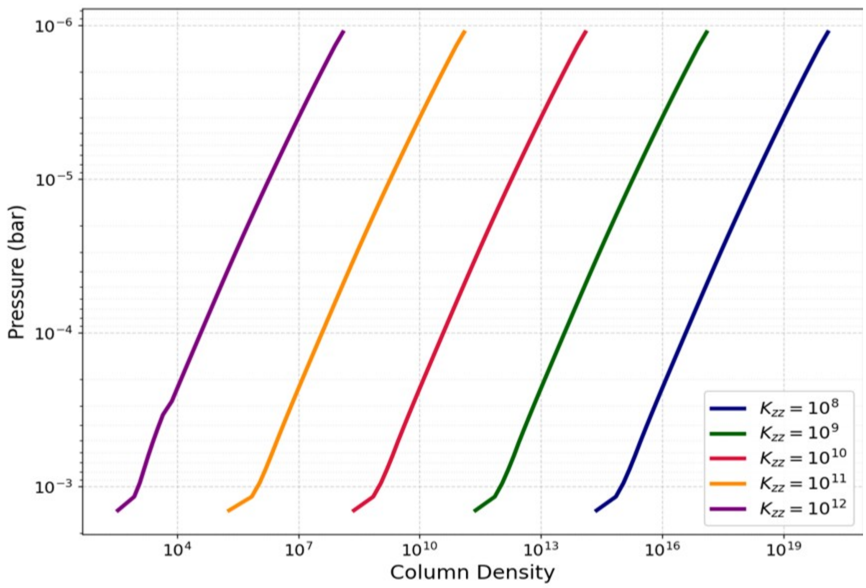

Figure 3.8: For varying K_{zz} we plot Pressure vs column density and seeing the relation.

- **Higher the value of K_{zz}, higher is the droplet radius** : K_{zz} controls w^* which controls the balance equation of v_{fall} and w^*, so if L is fixed for atmosphere increasing K_{zz} means increasing v_{fall} which is increased through increase in r_w
- **Higher the value of K_{zz}, lower is the column density** :r_w influences r_g, column density is $\frac{1}{r_g^3}$

3.2.2 Variable K_{zz}

I have calculated the K_{zz} using the formula given in the appendix and plotted Pressure vs $log_{10}q_c$ for f_{sed} = 1, 3, 10. The results exactly overlap to that of VIRGA. In this variable eddy diffusion parameter calculation the latent heat is taken into account the the equation of convective heat flux is also added.

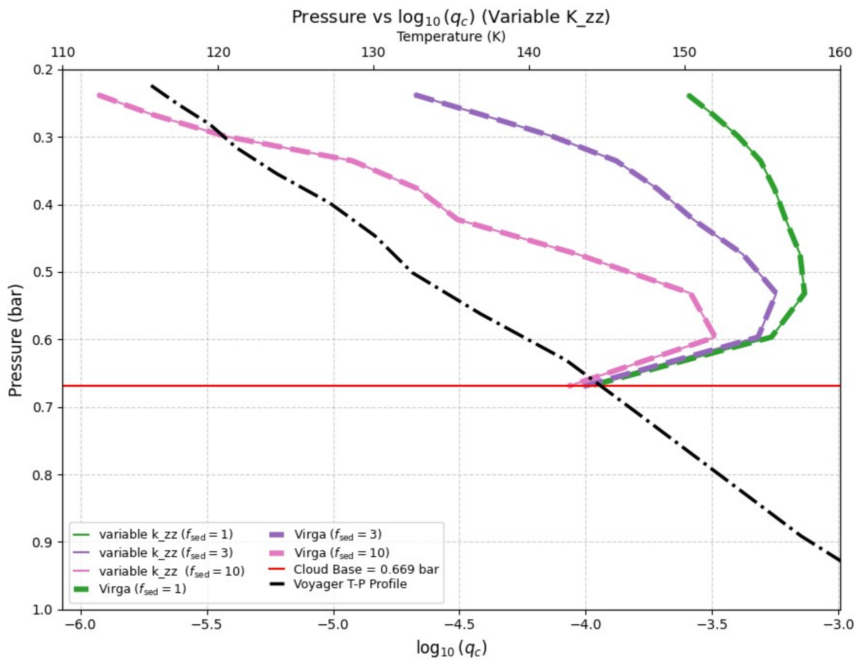

Figure 3.9: Plot of P vs $log_{10}q_c$ for f_{sed} = 1,3 10

3.3 Molecular weight and formation of clouds

The molecular weight of Al_2O_3 = **101.96 g/mol** and that of SiO_2= **60.08 g/mol** .

- The heavier the species, lower is the cloud base (closer to the surface)
- The clouds for the heavier species condenses and forms at higher pressure.

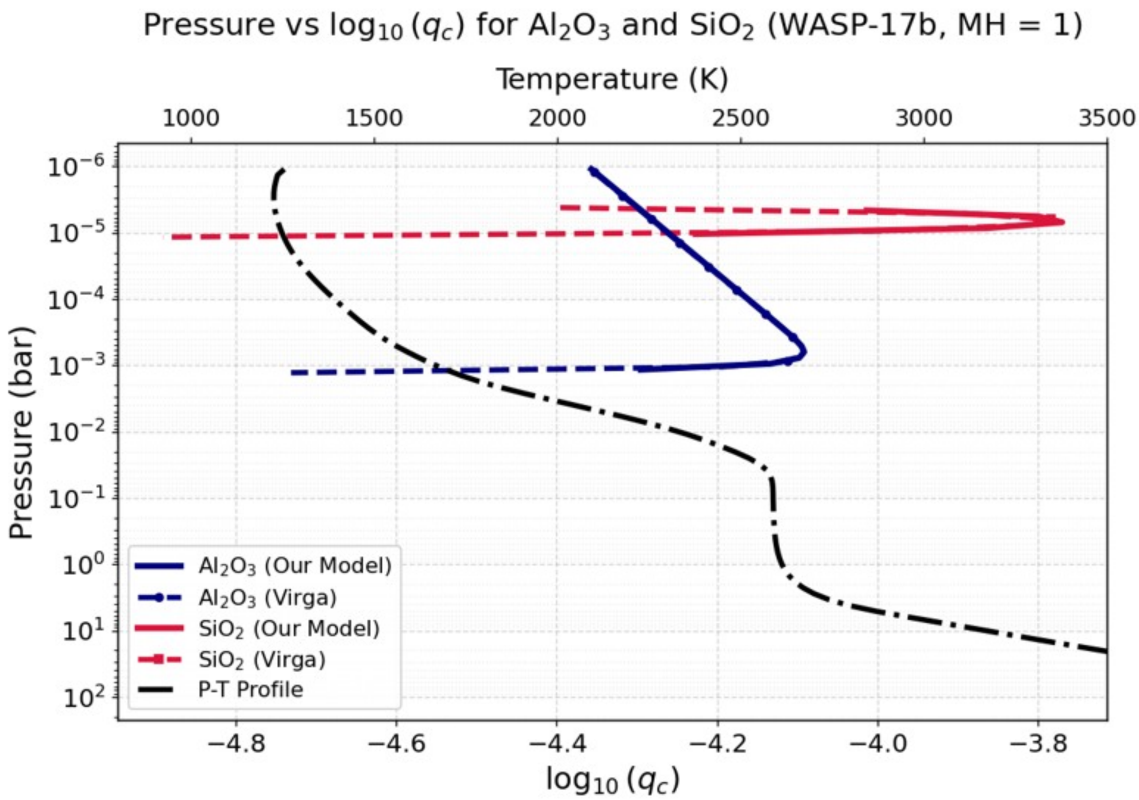

Figure 3.10: The species of Al_2O_3 and SiO_2 are plotted together in one frame

3.3.1 Metallicity and Cloud formation

Metallicity is the abundance of elements heavier than hydrogen and helium present in an astronomical object (stars and planets). More metallicity means more mass available for condensation.The higher the metallicity, the more is the cloud formation.

Higher metallicity leads to the shift to the cloud base at a higher pressure Clouds are formed at high temperature for higher metallicity

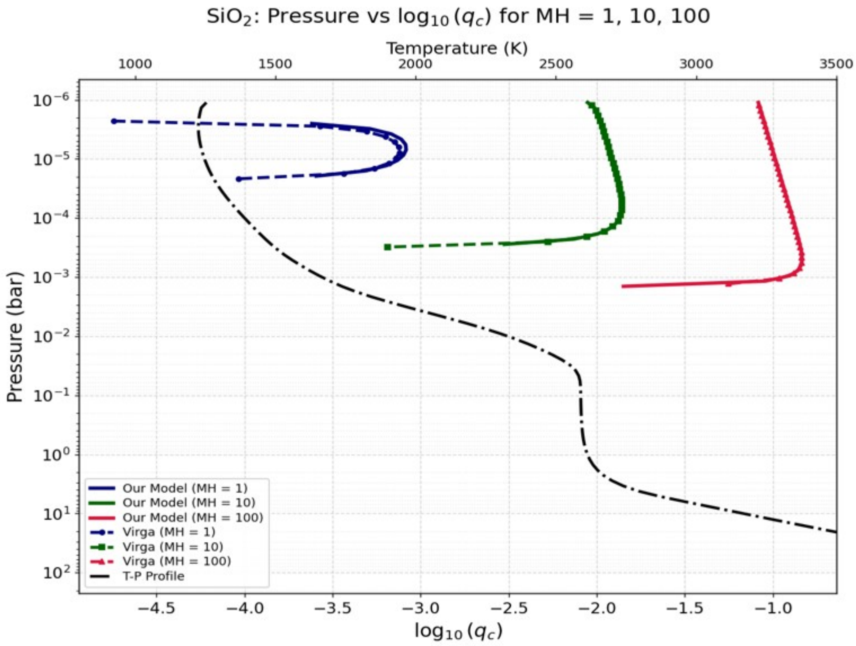

Figure 3.11: Plot of P vs Condensate Mass mixing ratio for various metallicity = 1, 10 ,100

- More metallicity means more q_c which means more condensable material. But in sedimenting cloud models, larger q_c with finite settling often tends to favor larger characteristic particle radii because: larger particles carry more mass efficiently. Higher the metallicity higher is the effective droplet radius.

- Metallicity adds so much more condensable material that even though particles become larger, the total integrated particle column still increases larger condensate loading more cloud mass larger integrated particle abundance. Higher the metallicity, higher is the Column density.

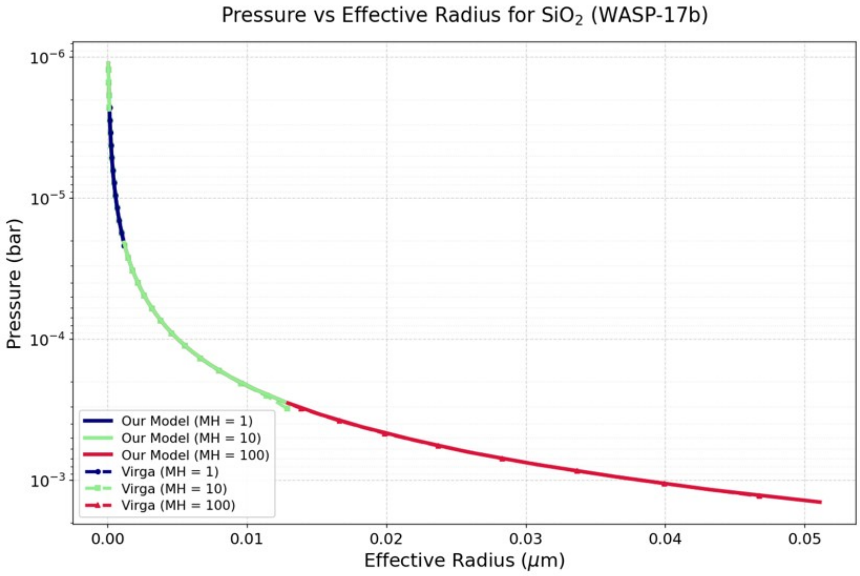

Figure 3.12: Plot of P vs Effective Radius for various metallicity = 1, 10 ,100

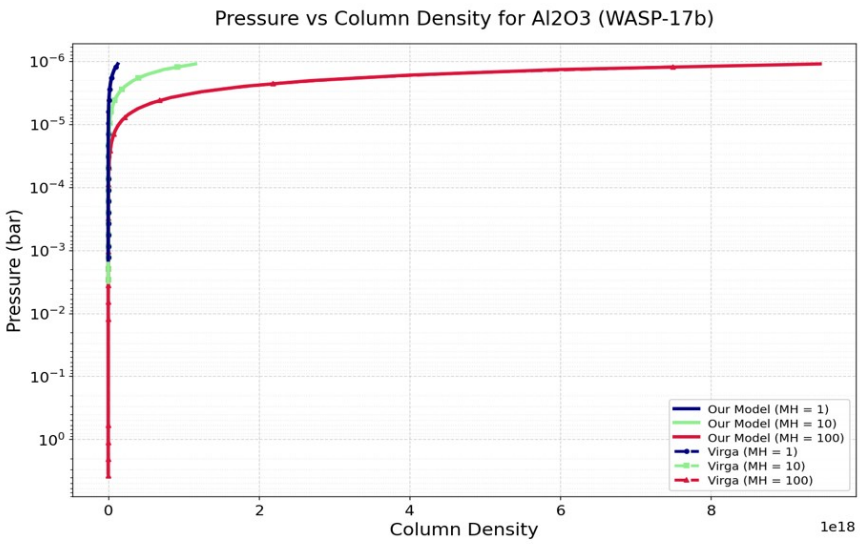

Figure 3.13: Plot of P vs Column Density for various metallicity = 1, 10 ,100

Chapter 4

Summary and Conclusions

In this semester, we made significant progress toward reconstructing the *Ackerman & Marley (2001)* cloud model for ammonia (NH_3) clouds in Jupiter's atmosphere. Although the Python implementation is not yet fully complete, the present version successfully captures the essential physical ingredients of the model and reproduces the expected qualitative behavior of Jovian cloud formation. As an important complementary step, we also developed a condensate selection and visualization framework in which the atmospheric pressure–temperature (P–T) profile can be compared directly against the saturation vapor pressure curves of multiple condensable species. This allows us to identify, for a chosen species, the approximate condensation level where the atmospheric profile intersects the corresponding saturation curve, thereby providing a simple and physically intuitive way to determine likely cloud-forming species in both Jovian and exoplanetary atmospheres. We have also started to compare our model with the more developed and newer cloud-physics model VIRGA.

Our implementation is based on the balance between upward turbulent mixing and downward sedimentation of condensate particles. In the governing differential equation, the first term represents diffusive transport driven by convective and turbulent mixing, while the second term accounts for sedimentation under gravity. Together, these processes, along with the particle fall radius r_w, determine the vertical condensate mixing ratio profile q_c. The resulting cloud structure is therefore controlled by the competition between atmospheric mixing and particle settling, parameterized by the sedimentation efficiency $f_{\rm rain}$ (or equivalently $f_{\rm sed}$).

The results obtained so far are qualitatively consistent with the theoretical expectations of the *Ackerman & Marley (2001)* framework. In particular, cloud bases form near the pressure–temperature level where the vapor mixing ratio equals the saturation mixing ratio ($q_v = q_s$), condensate mass increases above this level, and particle sizes show a strong dependence on f_{rain}. Larger sedimentation efficiencies produce larger particles, thinner and more compact cloud decks, and lower condensate abundances aloft, while smaller f_{rain} values lead to smaller particles and more vertically extended clouds.

At present we see that our model agrees well with the VIRGA model results and the discripencies of number density and effective radius can be explained with the fact that VIRGA is using a new scipy.optimize function called curve-fit, which is a better fit to the fall velocity model than the original equations of power law approximation method used by Ackerman and Marley. We see that we are able to match the exact cloud bases and patterns in case of Jupiter and WASP -17b our studies With the various metallicites and eddy diffusion parameters also give us important information about the formation of the clouds and can lead to some further important studies about the atmospheres of the exoplanets.

Overall, while further refinement is needed to improve the quantitative accuracy of the model, the present reconstruction already provides a physically consistent and promising framework for studying Jovian cloud microphysics. Beyond Jupiter, this work also forms the basis for extending the same methodology to exoplanetary atmospheres. In particular, the numerical tools developed here—including root-finding, particle fall-speed calculations, vertical transport, and condensate stability analysis through vapor-pressure curve comparisons—can be directly incorporated into our group's adaptive atmospheric model **SANSAR**. This will enable self-consistent cloud formation calculations not only for giant planets in the Solar System but also for hot and warm exoplanets such as WASP-17b, where condensates such as silicates may play a major role in shaping the observed spectra.

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Appendix A

Some basic definitions and formulas

- **Hydrostatic equilibrium**- $p(z) = \int_z^{\infty} g\rho dz$, where height is z,$\rho$ is the density of the air, g is the gravitational acceleration of the planet.
- **Dew point temperature** - the temperature to which surrounding atmosphere must be cooled, at constant pressure and moisture content, in order for it to become fully saturated with the condensate species.
- **Condensate** - a liquid that results from the cooling and condensation of a gaseous species
- q_c - moles of condensate per moles of atmosphere
- q_v - moles of vapor per moles of atmosphere
- q_t - sum of moles of (condensate+vapor) per moles of atmosphere
- q_s - moles of vapor per moles of atmosphere when atmosphere is saturated
- $R_d = (1000 \times \frac{8.3145}{2.2} JK^{-1}kg^{-1})$, where 8.3145 is the universal gas constant and 2.2 is the molecular weight of dry air in **Hydrogen-Helium** dominated atmosphere of Jupiter
- $R_v = (1000 \times \frac{8.3145}{18.016} JK^{-1}kg^{-1})$, where 8.3145 is the universal gas constant and 18.016 is the molecular weight of water
- **Convective Current** - a circulating flow within a fluid (liquid or gas) that occurs when differences in temperature cause corresponding differences in density of the particles.
- **Lapse Rate** (Γ_i) - the rate at which temperature changes with an increase in altitude.
- **Dry Adiabatic Lapse Rate** ($\Gamma_{adiabatic}$) - $\frac{g}{c_p}$, g is gravitational acceleration of the planet and c_p is the specific heat constant at constant pressure

- **Mixing length theory**- A common local first order closure of the turbulence closure problem. It is analogous to molecular diffusion, it assumes that flux is linearly proportional to and directed down the local gradient. $F_H = -K\frac{\delta\bar{\theta}}{dz}$

- **Eddy diffusion coefficient**- a parameter used instead of molecular diffusivity and is a constant of proportionality which increases with the intensity of the turbulence, it varies by height above the ground, mean wind shear and surface heating by the sun through radiation. Formula : [6]

$$K_{zz} = \frac{H}{3}(\frac{L}{H})^{\frac{4}{3}}(\frac{RF_{conv}}{\mu\rho_a c_p})^{\frac{1}{3}}$$

  where $H = RT/\mu g$, μ is atmospheric molecular weight, T is temperature, R is universal gas constant, L is turbulent mixing length, $F_conv = \sigma T_{eff}^4$ where σ is Stefhan - Boltzmann Constant ,T_{eff}^4 is the temperature at the surface of the planet, ρ_a is atmospheric density, c_p specific heat of the atmosphere.

- **Mixing Length** - The average size of the length of eddies formed during turbulent mixing length. Formula:

$$L = H \times \frac{\Gamma_i}{\Gamma_{adiabatic}}$$

  where H is the scale height

- **Convective Velocity scale (w_*)** - a velocity scale which is useful to characterize the turbulent mixing due to free convection in an unstable stratified atmosphere.

- **Root Finding Algorithm** - Iterative method for finding the roots in a continuous equation.

- **Brent's Method** - A compilation of three techniques to achieve the root of a function - secant method, bisection and inverse quadratic interpolation

- **Formula to calculate Saturation Vapor pressure (P_{sat}) for Ammonia** : $e_s = exp(10.53 - \frac{2161}{T} - \frac{86596}{T^2})$ where e_s is in $dynes\ cm^{-2}$

Appendix B

Some Derivations

B.1 Analytical derivation of the cloud formation equation

here $K_{ZZ} = K$

$$-K\frac{dq_t}{dz} - f_{sed}w_*q_c = 0 \tag{B.1}$$

$$-K\frac{dq_t}{dz} = f_{sed}w_*q_c \tag{B.2}$$

$$\frac{-K\frac{dq_t}{dz}}{q_c} = f_{sed}w_* \tag{B.3}$$

$$-K\int\frac{1}{q_c}\frac{dq_t}{dz}dz = zf_{sed}w_* \tag{B.4}$$

integrating it over altitude

$$-K\int_{L1}^{L2}\frac{dq_t}{q_t - q_v} = zf_{sed}w_* \tag{B.5}$$

$$-Klog(q_t - q_v)|_{L1}^{L2} = z|_{L1}^{L2}f_{sed}w_* \tag{B.6}$$

now we know : $w_* = K_{zz}/L$ putting it in the equation and changing the indexing of $q_t\&q_c$ for $L_1\&L_2$:

$$-K[log(q_{t,i} - q_{v,i}) - log(q_{below} - q_{v,i})] = \Delta zf_{sed}\frac{K}{L} \tag{B.7}$$

$$log\frac{(q_{t,i} - q_{v,i})}{(q_{below} - q_{v,i})} = -f_{sed}\frac{\Delta z}{L} \tag{B.8}$$

This is the analytical solution of the equation.

B.2 Similarity with other physical equations:

1st Term

Fick's 1st law of Diffusion is analogous to the first term of the equation of cloud model:

Fick's first law of diffusion:

$$J = -D\frac{dC}{dx} \tag{B.9}$$

J = diffusive flux (amount of substance crossing unit area per unit time)

D = diffusion coefficient

C = concentration of the substance

x = spatial coordinate

$-$ = negative sign shows flux goes down the concentration gradient

It is analogous to the first term of the equation $-K\frac{dq_t}{dz} - f_{rain}w_*q_c = 0$ in:

$$T_1 = -K\frac{dq_t}{dz} \tag{B.10}$$

where , K is analogous to D- it is the eddy diffusion coefficient

q_t is analogous to C - it is total mass mixing ratio

here x =z, which is the vertical coordinate

2nd Term

The second term is analogous to the sedimentation flux. Sedimentation flux is defined as : the rate at which sediment is transported across a specific area, such as an atmosphere. It is calculated by measuring the concentration of suspended sediment and the velocity of the fluid carrying it.

Therefore sedimentation flux will be:

$$F_{sed} = f_{rain} v_{fall} q_c \tag{B.11}$$

where :

q_c = condensate mass mixing ratio (g condensate / g air)

v_{fall} = particle terminal velocity (cm/s)

Now, in our equation $f_{rain} = \frac{v_{fall}}{w_*}$

Thus, $v_{fall} = f_{rain} \times w_*$ So, comparing to our second term of the main equation

$$f_{rain} w_* q_c = f_{rain} v_{fall} \tag{B.12}$$

Similar to real life physical equation

our equation

$$-K\frac{dq_t}{dz} - f_{rain} w_* q_c = 0$$

is similar to charged particle transport.

$$-D\frac{dn}{dx} + \mu n E$$

where D is charged diffusivity, μ mobility of ions, n is the concentration, E is the electric field, x is the spatial coordinate.

B.3 Reynolds number and Droplet size distribution

In the atmosphere apart from turbulent flow there is also laminar flow which depends on the temperature, density of atmospheric (which is assumed to be uniform in this case) mass and diameter of the particle in consideration along with velocity of the particle falling. Reynolds number comes here in play as it tells us whether the particle is going through laminar flow or dynamic flow.

$$R_e = \frac{\rho v L}{\eta} \tag{B.13}$$

where, ρ - density, v - speed of the particle, L - characteristic length scale, η - dynamic viscosity.

$$\eta = \frac{5}{16} \frac{\sqrt{\pi m k_B T}(k_B T/\epsilon)^{0.16}}{\pi d^2 \times 1.22} \tag{B.14}$$

here, d is molecular diameter, k_B is Boltzmann constant, m is the mass of the particles,ϵ is the depth of the Leonard-Jones potential well for the atmosphere [1]

In our model we have defined the Reynolds number as :

$$2r\frac{\rho_a v_{fall}}{\eta} \tag{B.15}$$

here, r is the radius of the droplet, v_{fall} is calculated according to the distinction of the Reynolds number'

For Reynolds number less than 1

Over here we have laminar flow;

$$v_{fall} = \frac{2\beta g r^2 \Delta\rho}{9\eta} \tag{B.16}$$

here, $\beta = (1 + 1.26K_N)$ accounts for gas kinetic effects is the Cunningham correction factor for non-continuum effects, in which Knudsen Number N_{KN} is the ratio of the molecular free path to the droplet radius $N_{KN} = \frac{\lambda}{r}$ and $\delta\rho$ is difference of densities of the condensate and the atmosphere.

For Reynolds number between 1 and 1000

For Reynolds number in this category we have intermediate turbulent flow;

$$C_d R_e^2 = \frac{32r^3\rho_a(\Delta\rho)}{3\eta^2} \tag{B.17}$$

hereC_d is the drag coefficient,which is used in drag force

$$F_\eta = \frac{C_d}{2}\pi\rho_a r^2 v_f^2$$

[5] Now, $C_d R_e^2$ is independent of the fall velocity and hence we fit $y = log(R_e)$ as a function of $x = log(C_d R_e^2)$. The fit goes as $y = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + b_4 x^4 + b_5 x^5 + b_6 x^6$ where $b_0, b_1, b_2, b_3, b_4, b_5, b_6$ are coefficients of the fit.

For Reynolds number greater than 1000

Here the flow i completely turbulent. So, we assume the drag coefficient to have a fixed asymptotic value $C_d = 0.45$ [1] which gives us

$$v_{fall} = \beta\sqrt{\frac{8gr\Delta\rho}{3C_d\rho_a}} \tag{B.18}$$

B.4 Calculation of Effective droplet radius

The geometric mean radius r_g via the equation 2.7:

with the associated column droplet number concentration (cm^{-2}) via the equation 2.8 :

Then , the effective droplet radius is calculated as :

$$r_{eff,i} = r_{g,i}exp(\frac{5}{2}ln^2\sigma_g) \tag{B.19}$$

here $\sigma_g = 2$, is the geometric standard deviation, it can lie between 1.1 - 2.5

B.5 Calculation of Number Density

$$Ndz_i = \frac{3\,\rho_{a,i}\,q_{c,i}}{4\pi\rho_p\,r_{g,i}^3}\exp\left(-\frac{9}{2}\ln^2\sigma_g\right)\,dz_i \tag{B.20}$$